JEE APEX FTS – JEE ADV. FULL SYLLABUS TEST-08
[2023 - P2]
KEY SHEET

MATHEMATICS

1	**B**	2	**A**	3	**C**	4	**D**	5	**BD**	6	**ABC**
7	**BD**	8	**4**	9	**8**	10	**28**	11	**5**	12	**2**
13	**12**	14	**0.40**	15	**16**	16	**3**	17	**16.50**		

PHYSICS

18	**A**	19	**C**	20	**C**	21	**A**	22	**BC**	23	**BC**
24	**BC**	25	**2**	26	**8**	27	**8**	28	**4**	29	**7**
30	**5**	31	**90**	32	**12**	33	**4**	34	**8**		

CHEMISTRY

35	**C**	36	**C**	37	**C**	38	**A**	39	**AC**	40	**ACD**
41	**BD**	42	**2**	43	**5**	44	**7**	45	**3**	46	**6**
47	**2**	48	**394**	49	**3272**	50	**5**	51	**8**		

SOLUTIONS
MATHEMATICS

1. $$f(n)=n+\sum_{r=1}^{n}\frac{16r+(9-4r)n-3n^2}{4rn+3n^2}$$

$$f(n)=n+\sum_{r=1}^{n}\frac{(16r+9n)-(4rn+3n^2)}{4rn+3n^2}$$

$$f(n)=n+\left(\sum_{r=1}^{n}\frac{16r+9n}{4rn+3n^2}\right)-n=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{\left(16\left(\frac{r}{n}\right)+9\right)\frac{1}{n}}{4\left(\frac{r}{n}\right)+3}$$

$$= \int_0^1 \frac{16x+9}{4x+3}dx = \int_0^1 4\,dx - \int_0^1 \frac{3\,dx}{4x+3} = 4 - \frac{3}{4}\left(\ln|4x+3|\right)_0^1 = 4 - \frac{3}{4}\ln\frac{7}{3}$$

2. $g(x) = (x-a)(x-b)(x-c)$

$f(x) = g(x)h(x)$

$x^4 + x^3 + bx^2 + 100x + c = \left(x^3 + ax^2 + x + 10\right)(x+r)$

$x^4 + x^3 + bx^2 + 100x + c - \left(x^3 + ax^2 + x + 10\right)(x+r) = 0$

$x^3 : 1 - (a+r) = 0$ and $x : 100 - (10+r) = 0 \Rightarrow r = 90,\ a = -89$

$g(x) = x^3 - 89x^2 + x + 10$

$g(1) = 1 + a + 1 + 10 = 1 - 89 + 1 + 10 = -77$

$h(1) = 1 + r = 1 + 90 = 91$

$f(1) = g(1)h(1) = -77.91 = -7007$

3. $a^3 + b^3 + c^3 = (a+b+c)^3 = a^3 + b^3 + c^3 + 3(a+b)(b+c)(c+c)$

$\Rightarrow (a+b)(b+c)(c+a) = 0$

$(\tan x + \cot x)(\tan x + 2)(\cot x + 2) = 0$

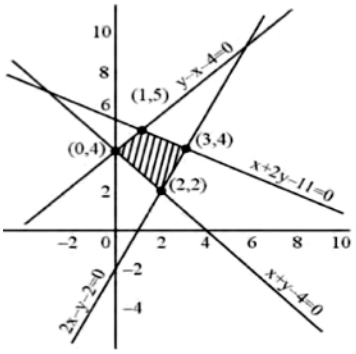
$\tan x + \cot x = 0$ (or) $\tan x + 2 = 0$ (or) $\cot x + 2 = 0$

$\tan^2 x = -1$ (or) $\tan x = -2$ (or) $\cot x = -2$

No solution | 4 solutions | 4 solutions

Left | Right

4. For $R_1 : (y - x - 4)(y + x - 4) \le 0$



$R_2 : (x + y - 4)(y - 2x + 2) \ge 0$

$R_3 : (2x - y - 2)(x + 2y - 11) \ge 0$

$$\text{Area} = \frac{1}{2}\begin{pmatrix} 0 & 2 & 3 & 1 & 0 \\ 4 & 2 & 4 & 5 & 4 \end{pmatrix} = \frac{1}{2}(8 + 15 + 4 - 8 - 6 - 4) = 4.5$$

5. $\vec{a} - = 3\hat{i} + \hat{j} - \hat{k}$

$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$$\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3 - c_1 \\ 1 - c_2 \\ -1 - c_3 \end{pmatrix}$$

Multiply and compare

$b_2c_3 - b_3c_2 = c_1 - 3.....(1)$

$c_3 - b_3c_1 = 1 - c_2.....(2)$

$c_2 - b_2c_1 = 1 + c_3.....(3)$

$\vec{b} \times \vec{c} = \vec{c} - \vec{a}$ -

Take dot product with $\vec{b}$

$0 = \vec{c}.\vec{b} - \vec{a}.\vec{b}$ -

$\vec{b}.\vec{c} = 0$

$\vec{b} \perp \vec{c}$

$\vec{b}\hat{}\vec{c} = 90^0$

Take dot product with $\vec{c}$

$0 = |\vec{c}|^2 - \vec{a}.\vec{c}$

$\vec{a}.\vec{c} = |\vec{c}|^{-2}$

$\vec{a}.\vec{c} \neq 0$

$\vec{b} \times \vec{c} - = \vec{c} - \vec{a}$

Squaring

$|\vec{b}|^2 \ |\vec{c}|^2 = |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c}.\vec{a}$

$|\vec{b}|^2 \ |\vec{c}|^2 = |\vec{c}|^2 + 11 - 2|\vec{c}|^2$ -

$|\vec{b}|^2 \ |\vec{c}|^2 = 11 - |\vec{c}|^2$

$|\vec{c}|^2 = \dfrac{11}{|\vec{b}|^2 + 1}$

$|\vec{c}| \leq \sqrt{11}$

$b_2 - b_3 = -3$ also

$b_2^2 + b_3^2 - 2b_2b_3 = 9 \quad b_2b_3 > 0$

$b_2^2 + b_3^2 = 9 + 2b_2b_3$

$b_2^2 + b_3^2 = 9 + 2b_2b_3 > 9$

$b_2^2 + b_3^2 > 9$

$|\vec{b}| = \sqrt{1 + b_2^2 + b_3^2}$

$|\vec{b}| > \sqrt{10}$

6. $a_1 = 7, \ d = 8$

$T_{n+1} - T_n = a_n \ \forall n \geq 1$

$S_n = T_1 + T_2 + T_3 + ... + T_{n-1} + T_n$

$S_n = \ T_1 + T_2 + T_3 + ... + T_{n-1} + T_n$

On subtracting

$T_n = T_1 + a_1 + a_2 + ... + a_{n-1}$

$T_n = 3 + (n-1)(4n-1)$

$T_n = 4n^2 - 5n + 4$

$$\sum_{k=1}^{n} T_k = 4\sum n^2 - 5\sum n + 4n$$

$$T_{20} = 1540$$

$$T_{30} = 3454$$

$$\sum_{k=1}^{30} T_k = 35615$$

$$\sum_{k=1}^{20} T_k = 10510$$

7. $P_1 : x + 3y + 5z = 2,\ P_2 : x - 2y + z = 5$

P_1 and P_2 are perpendicular

If directional ratios of line represented by $P_1 = 0,\ P_2 = 0$ are (a, b, c)

$$\Rightarrow a + 3b + 5c = 0,\ a - 2b + c = 0 \Rightarrow \frac{a}{13} = \frac{b}{4} = \frac{c}{-5}$$

dr's may be (13, 4, -5)

Option A is wrong

Foot of $\perp$' from P to P_1 is (x_2, y_2, z_2)

$$\frac{x_2 + 4}{1} = \frac{y_2 - 7}{3} = \frac{z_2 - 11}{5} = \left(\frac{-4 + 21 + 55 - 2}{35}\right) = -2 \Rightarrow x_2 = -6,\ y_2 = 1,\ z_2 = 1$$

If foot of perpendicular from P to P_3 is (x_3, y_3, z_3)

$$\frac{x_3 + 4}{1} = \frac{y_3 - 7}{-2} = \frac{z_3 - 11}{1} = -\left(\frac{-4 - 14 + 11 - 5}{6}\right) = 2 \Rightarrow x_3 = -2,\ y_3 = 3,\ z_3 = 13$$

$$Q = (-6, 1, 1),\ R = (-2, 3, 13)$$

Equation of plane $(13x + 4y - 5z + 79) = 0$

8. $$a \in \left(0, \frac{\pi}{4}\right),\ \beta \in \left(-\frac{\pi}{4}, 0\right) \Rightarrow \alpha + \beta \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$$

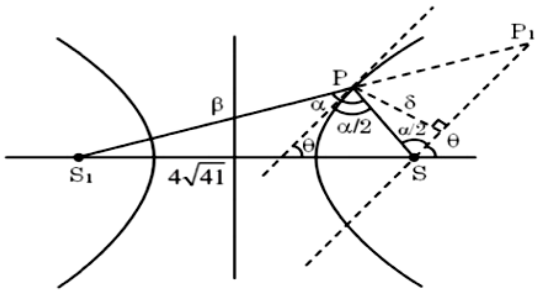
$$\sin(\alpha + \beta) = \frac{1}{3},\ \cos(\alpha - \beta) = \frac{2}{3}$$

$$\left(\frac{\sin\alpha}{\cos\beta} + \frac{\cos\alpha}{\sin\beta} + \frac{\cos\beta}{\sin\alpha} + \frac{\sin\beta}{\cos\alpha}\right)^2 = 4\cos^2(\alpha - \beta)\left(\frac{1}{\sin 2\beta} + \frac{1}{\sin 2\beta}\right)^2$$

$$= 4\cos^2(\alpha - \beta)\left(\frac{2\sin(\alpha + \beta)\cos(\alpha - \beta)}{\sin 2\alpha\ \sin 2\beta}\right).....(1)$$

$$= \frac{16\cos^4(\alpha - \beta)\sin^2(\alpha + \beta) \times 4}{\left(\cos 2(\alpha - \beta) - \cos 2(\alpha + \beta)\right)^2} = 64 \times \frac{16}{81} \times \frac{1}{9} \frac{1}{\left(2 \times \frac{4}{9} - 1 - 1 + \frac{2}{9}\right)^2} = \frac{64 \times 16}{81 \times 9} \cdot \frac{81}{64} = \frac{16}{9}$$

9.



$S_1P = SP = 20$

$$\beta - \frac{\delta}{\sin\frac{\alpha}{2}} = 20$$

$$\beta^2 + \frac{\delta^2}{\sin^2\frac{\alpha}{2}} - 400 = \frac{2\beta\delta}{\sin\frac{\alpha}{2}}$$

$$\frac{1}{SP} = \frac{\sin\frac{\alpha}{2}}{\delta}$$

$$\frac{\lambda - 128}{\lambda} = \cos\alpha$$

$$\lambda(1 - \cos\alpha) = 128$$

$$\frac{\beta\delta}{\sin\frac{\alpha}{2}} . 2\sin^2\frac{\alpha}{2} = 128 \qquad \beta\delta\sin\frac{\alpha}{2} = 64$$

10. $\alpha + \beta = -z_1,\ \alpha\beta = z_2 + m$

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = z_1^2 - 4(z_2 + m) = z_1^2 - 4z_2 - 4m = 16 + 20i - 4m$$

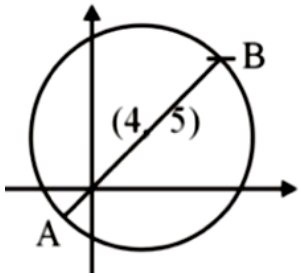
$$(\alpha - \beta)^2 = 4(4 + 5i - m)$$

$$|\alpha - \beta|^2 = 4\,|4 + 5i - m|$$

$$4 \times 7 = 4\,|4 + 5i - m|$$

$$|m - (4 + 5i)| = 7$$

So ‘m’ lies on circle centre (4, 5) and radius 7.



$$|M|_{max} = \sqrt{4^2 + 5^2} + 7 = 7 + \sqrt{41}$$

$$|M|_{min} = \left|\sqrt{4^2 + 5^2} - 7\right| = 7 - \sqrt{41}$$

Their sum is $\left(7 + \sqrt{41}\right) + \left(7 - \sqrt{41}\right) \qquad \lambda = 14$

$$\frac{\lambda}{5}=\frac{14}{5}=2.80$$

11. $$\frac{\sum_{i=1}^{20}(x_i+i)}{20}=15.5$$

$$\frac{\sum_{i=1}^{20}x_i}{20}+\frac{\sum_{i=1}^{20}i}{20}=15.5$$

$$\overline{x}+\frac{20\times 21}{2\times 20}=15.5$$

$$\overline{x}=15.5-10.5=5\ .....(1)$$

$$\sum_{i=1}^{20}(x_i-5)^2=100$$

$$\sum x_i^2+25\times 20-10\times 5\times 20=100$$

$$\sum x_i^2=1000+1000-500=600$$

$$SD=\sqrt{\frac{\sum x_i^2}{20}-(\overline{x})^2}=\sqrt{\frac{600}{20}-(5)^2}=\sqrt{30-25}=\sqrt{5}\Rightarrow \text{var}(x_i-2)=\text{var}(x_i)=5$$

12. $$P(E)=\frac{1}{2}\ P(F_k)={}^{n}C_k\times\frac{1}{2^n}$$

$$P(E\cap F_k)={}^{n-1}C_{k-1}\times\left(\frac{1}{2}\right)^n$$

$$P(E\cap F_k)=P(E)\times P(F_k)$$

$${}^{n-1}C_{k-1}\times\left(\frac{1}{2}\right)^n=\left(\frac{1}{2}\right)^n\times\frac{1}{2}\times {}^{n}C_k N \qquad \text{or } 2\times {}^{n-1}C_{k-1}={}^{n}C_k$$

$$\frac{{}^{n}C_k}{{}^{n-1}C_{k-1}}=2=\frac{n}{k}$$

13. $(a-b-c)=0,\Rightarrow a=b+c\Rightarrow a=2,\ b=c=1$

So, both the roots of the equation $x^2+2(p-(3))x+9=0$ lies in $(-(2+1+3),1)$

i) $\Delta\geq 0\Rightarrow P\leq 0(\text{or})\ 6\leq P$

ii) $f(-6)>0\Rightarrow\frac{27}{4}>P$ and iii) $f(1)>0\Rightarrow P>2$

Therefore common region of (i), (ii), (iii) $\Rightarrow 6\leq p<\frac{27}{4}\Rightarrow p=6$

If a, g_1, g_2,....g_{20}, p are in G.P $\Rightarrow 2g_1$, g_2,....., g_{20}, 6 are in G.P $\Rightarrow g_4\ g_{17}=12$

14. $P_1: x+y+z=a$

$P_2: x+5y-z=b$

$P_3: x-y+2z=c+2$

For consistent $P_2=3P_1-2P_3\Rightarrow b=3a-2c-4$

$\Rightarrow 3a-b-2c-4=0 \qquad$ so $P: 3x-y-2z-4=0$

$$D=\frac{4}{\sqrt{14}}$$

$$M=\begin{vmatrix} a & -2 & b \\ 2 & 1 & c \\ 1 & 0 & 3 \end{vmatrix} \quad =3a-2(c-6)+b(-1) \quad =3a-b-2c+12 \quad =4+12=16$$

15. $$S_1=\int_{\frac{\pi}{16}}^{\frac{7\pi}{16}} f(x)dx=\int_{\frac{\pi}{16}}^{\frac{7\pi}{16}} \sin^2 x\, dx \ldots\ldots\ldots(1)$$

$$S_1=\int_{\frac{\pi}{16}}^{\frac{7\pi}{16}} \sin^2\left(\frac{\pi}{16}+\frac{7\pi}{16}-x\right)dx \quad =\int_{\frac{\pi}{16}}^{\frac{7\pi}{16}} \cos^2 x\, dx \ldots\ldots\ldots(2)$$

By 1 + 2

$$2S_1=\int_{\frac{\pi}{16}}^{\frac{7\pi}{16}} \left(\sin^2 x+\cos^2 x\right)$$

$$2S_1=\frac{7\pi}{16}-\frac{\pi}{16}=\frac{3\pi}{8}$$

$$S_1=\frac{3\pi}{16} \qquad \Rightarrow \frac{16S_1}{\pi}=3$$

16. $P_1: x+y+z=a$

$P_2: x+5y-z=b$

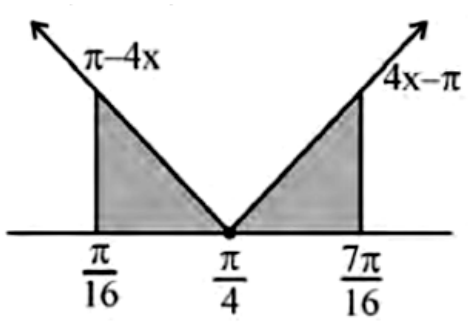
$P_3: x-y+2z=c+2$

For consistent $\qquad P_3=3P_1-2P_2 \Rightarrow b=3a-2c-4 \Rightarrow 3a-b-2c-4=0$

$$D=\frac{4}{\sqrt{14}}$$

$$M=\begin{vmatrix} a & -2 & b \\ 2 & 1 & c \\ 1 & 0 & 3 \end{vmatrix} \quad =3a-2(c-6)+b(-1) \quad =3a-b-2c+12 \quad =4+12=16$$

17.



$$S_2 = \int_{\frac{\pi}{16}}^{\frac{7\pi}{16}} f(x) g_2(x)\,dx = \int_{\frac{\pi}{16}}^{\frac{7\pi}{16}} \sin^2 x\,|4x-\pi|\,dx \ldots\ldots (1) \;= \int_{\frac{\pi}{16}}^{\frac{7\pi}{16}} \sin^2\left(\frac{\pi}{2}-x\right)\left|4\left(\frac{\pi}{2}-x\right)-\pi\right| dx$$

$$S_2 = \int_{\frac{\pi}{16}}^{\frac{7\pi}{16}} \left(\cos^2 x\right)|\pi - 4x|\,dx \ldots\ldots (2)$$

By 1 + 2 $\qquad\qquad \Rightarrow 2S_2 = \int_{\frac{\pi}{16}}^{\frac{7\pi}{16}} |4x-\pi|\left(\sin^2 x + \cos^2 x\right)dx$

$$2S_2 = \frac{33\pi^2}{64}$$

$$\frac{64S_2}{\pi^2} = \frac{33}{2} = 16.5$$

18. $\frac{1}{v}+\frac{1}{-600}=\frac{1}{20}$

or $\frac{1}{v}=\frac{31}{600}$ or $v=\frac{600}{31}\text{cm}=19.35\text{ cm}$

19. $d_1=\frac{20}{\frac{12}{6}-1}=20\text{cm}=0.2\text{ m}$

20. $\frac{1}{2}mv^2=vq\Rightarrow v=\sqrt{\frac{2qv}{m}}$

$D=\frac{2\left(2\times10^7\right)}{1.76\times10^2\times1.57\times10^{11}}=1.45\times10^{-3}\text{mm}$

$v_p:v_d:v_a=\sqrt{\frac{1}{1}}:\sqrt{\frac{1}{2}}:\sqrt{\frac{2}{4}}\qquad=1:\frac{1}{\sqrt{2}}:\frac{1}{\sqrt{2}}=\sqrt{2}:1:1$

21. Given Mass (m) = 0.4 kg. Its frequency (n) = 2 rev/sec.
Radius (r) = 1.2 m. We know that linear velocity of the body
$v=\omega r=\left(2\pi n\right)r=2\times3.14\times1.2\times2=15.08\text{ m/s}$

Therefore tension in the string when the body is at the top of the circle $\left(T\right)=\frac{mv^2}{r}-mg$

$=\frac{0.4\times\left(15.08\right)}{1.2}-\left(0.4\times9.8\right)=75.80-3.92=71.88\text{N}$

22. At $t=\frac{4}{3}\text{s},\ Y_1=\frac{5}{\left(3x-\frac{16}{3}\right)^2+2}$ and $Y_2=\frac{-5}{\left(3x-\frac{2}{3}\right)^2+2}$

$Y_1+Y_2\neq0$

At $t=\frac{3}{4}\text{s},\ Y_1=\frac{5}{\left(3x-3\right)^2+2}$ and $Y_2=\frac{-5}{\left(3x-3\right)^2+2}$

$Y_1+Y_2=0$

At $x=0,\ Y_1=\frac{5}{\left(3-4t^2\right)^2+2}$ and $Y_2=\frac{-5}{\left(-3+4t^2\right)+2}$

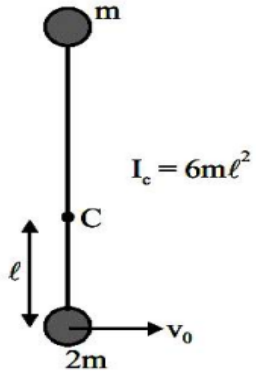
$Y_1+Y_2=0$

23. $\Delta A=\left(\frac{\Delta l}{l}+\frac{\Delta b}{b}\right)=A\pm\left(\frac{0.1}{10.1}+\frac{0.01}{1.00}\right)\times\left(10.0\times1.00\right)\text{cm}^2$

$=\pm0.02\times10=\pm0.2\text{ cm}^2$

24. For (B) : $2mv_0l=6ml^2\omega$

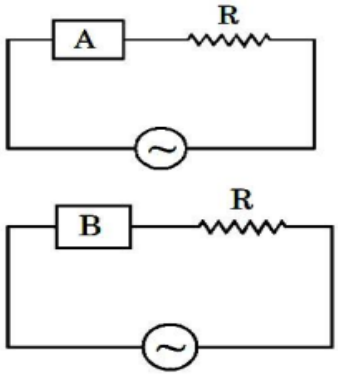
$\therefore\omega=\frac{v_0}{3l}$



$$T = 2m\omega^2 l = \frac{2mv_0^2}{9l}$$

for (C): $I_A = 2m(3l)^2$; $I_B = m(3l)^2$

25.



$\phi_1 = -30^0$

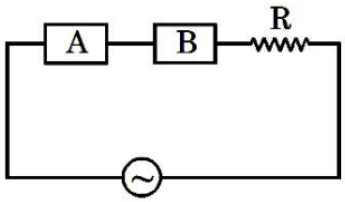
$\phi_2 = +60^0$

$$\tan\phi = \frac{X_L}{R}$$

$$\tan\phi_2 = \frac{X_C}{R}$$

$$X_L = \frac{R}{\sqrt{3}}$$

$$X_C = R\sqrt{3}$$



$$\tan\theta = \frac{(X_C - X_L)}{R} = \frac{\frac{R\sqrt{3} - R}{\sqrt{3}}}{R} = \frac{2}{\sqrt{3}}$$

26. $$R_{AB} = \int_0^1 \frac{\rho_0 X dx}{A} \quad = \int_0^1 \frac{24\pi X dx}{A} = 24 \times \frac{2}{2}\Big|_0^1 = 12\ \Omega$$

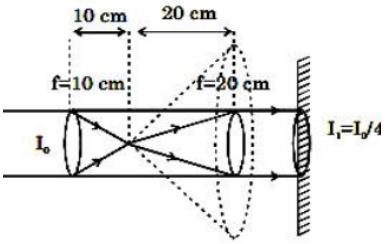
$$R_{AG}=\int_0^{\sqrt{\frac{2}{3}}}\frac{24\pi X\,dx}{\pi}=\frac{24}{2}\left[X^2\right]_0^{\sqrt{\frac{2}{3}}}=12\times\frac{2}{3}=8\Omega$$

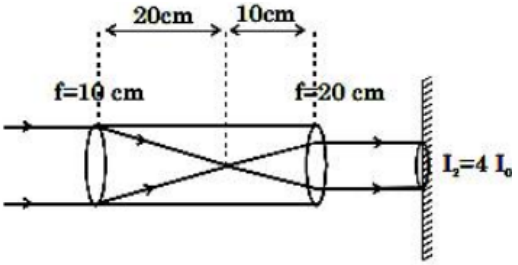
$$i_{in}\ \text{wire}=\frac{15}{3+12}=1A \qquad V_{AG}=1\times 8=8V$$

27. $\text{Slope}=-\frac{1}{2}$ $\qquad \ln\left(\frac{dN}{dt}\right)=-\frac{1}{2}$

$N=N_0e^{-\frac{1}{2}t}$, $\qquad$ After 4.16 years $\qquad \frac{N}{N_0}=e^{-(2.08)}=\frac{1}{8}$

28.





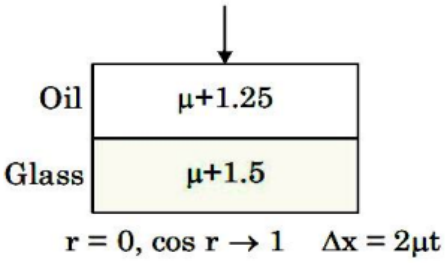
$$\therefore\frac{I_2}{I_1}=16;\ \therefore N=4$$

29. $$-\frac{1}{A}\frac{dQ}{dt}=\sigma\left(T^4-O^4\right)\Rightarrow -mc\frac{dT}{dt}=\sigma AT^4\Rightarrow T^{-4}\,dT=-\frac{\sigma A}{mc}dt$$

$$=\int_{200K}^{100K}T^{-4}\,dt=-\frac{\sigma A}{mc}\int_0^t dt\Rightarrow\frac{T^{-3}}{-3}\Big|_{200}^{100}=-\frac{\sigma A}{mc}t$$

$$\Rightarrow\frac{1}{(100)^3}-\frac{1}{(200)^3}=\frac{\sigma\left(4\pi r^2\right)}{\frac{4}{3}\pi r^3\rho c}\Rightarrow t=\frac{7}{72}\frac{r\rho c}{\sigma}\times 10^{-6}s\Rightarrow t=\frac{7}{72}\frac{\rho pc}{\sigma}\ \mu s$$

30. Optical path difference for the light transmitted through oil is $\Delta x=2\mu l\cos r$ For normal incidence



r = 0, cos r → 1 $\quad \Delta x = 2\mu t$

But the interface between oil and glass will produce an additional path difference of $\frac{\lambda}{2}$.

Therefore effective path difference.

$$\Delta x=2\mu t+\frac{\lambda}{2}$$

For constructive interference in transmitted light $2\mu t+\frac{\lambda}{2}=n\lambda,\ n=1,2,....$or

$$2\mu t=(2n-1)\frac{\lambda}{2}$$

for $n = 1, \lambda = 5\times10^{-6}\,m = 5000$ nm

for $n = 2, \lambda = \frac{5\times10^{-6}}{3}\,m = 1666.67$ nm

for $n = 3, \lambda = \frac{5\times10^{-6}}{5}\,m = 1000$ nm

for $n = 4, \lambda = \frac{5\times10^{-6}}{7}\,m = 714.29$ nm

for $n = 5, \lambda = \frac{5\times10^{-6}}{5}\,m = 555.55$ nm

for $n = 6, \lambda = \frac{5\times10^{-6}}{11}\,m = 454.54$ nm

The wavelength which are strongly transmitted in visible range are : 714.29 nm, 555.55 nm and 454.54 nm.

31. Speed of train

$$v = \left(\frac{75}{60}\text{click / sec}\right)(20\text{m})\left(\frac{18}{5}\right) = 90\text{ km / hr}$$

32. $\Delta L = L\alpha\Delta T = (20\text{m})\left(12\times10^{-16}\right)(50) = 12$ mm

33. In ground state (n = 1) according to Bohr's theory :

$$mvR = \frac{h}{2\pi} \Rightarrow v = \frac{h}{2\pi mR}$$

Now, time period, $T = \frac{2\pi R}{v} = \frac{2\pi R}{\frac{h}{2\pi mR}} = \frac{4\pi^2 mR^2}{h}$

Magnetic moment, M = iA

where $i = \frac{\text{charge circulating}}{\text{time period of one revolution}} = \frac{e}{\frac{4\pi^2 mR^2}{h}} = \frac{eh}{4\pi^2 mR^2}$ and $A = \pi R^2$

$$\Rightarrow M = \left(\pi R^2\right)\left(\frac{eh}{4\pi^2 mR^2}\right) \Rightarrow M = \frac{eh}{4\pi m}$$

Direction of magnetic moment $\vec{M}$ is perpendicular to the plane of orbit

34. $\vec{\tau} = \vec{M}\times\vec{B}$

$\vec{\tau} = MB\sin\theta$

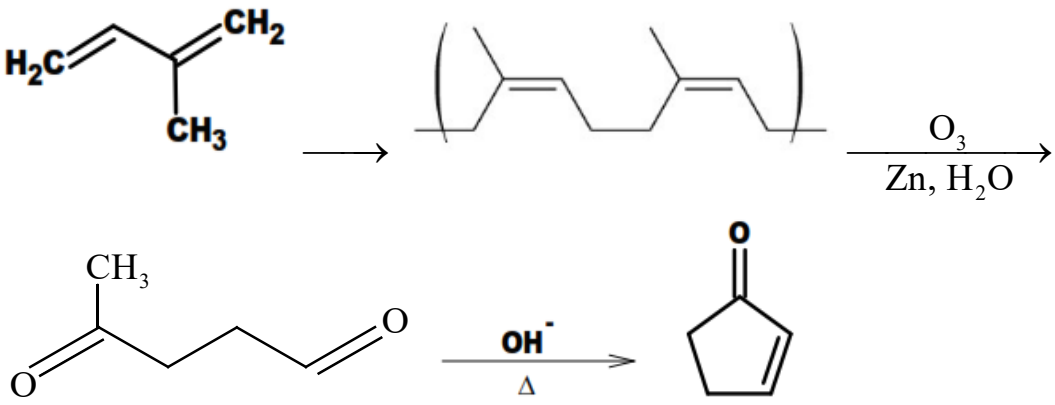
Where θ is the angle between $\vec{M}$ and $\vec{B}$

$\theta = 30^0$ $\qquad\qquad$ $\therefore \tau = \left(\frac{eh}{4\pi m}\right)(B)\sin 30^0 = \frac{ehB}{8\pi m}$

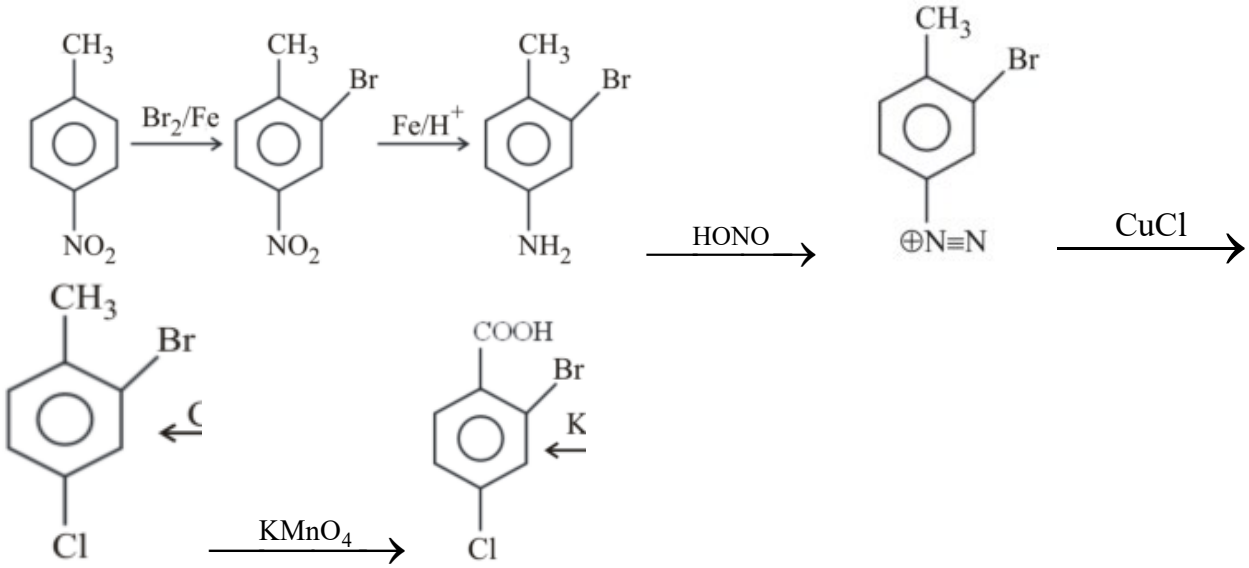
The direction of $\vec{\tau}$ is perpendicular to both $\vec{M}$ and $\vec{B}$

35. It is negative sol.

36.



37.



38. Bond angle is less than 90^0

39. Conceptual

40. ACD are correct

41. ABD are correct

42. $P_{gas} = K_H.X_{gas}$

$P_{gas} = 10 \times 0.8 = 8$ atm, $8 = 2.22 \times 10^4 X_{gas}$

$$X_{gas} = \frac{8}{2.22 \times 10^4} = 3.6 \times 10^{-4}, \frac{n_{gas}}{n_{water}} = 3.6 \times 10^{-4}$$

$n_{gas} = 3.6 \times 10^{-4} \times 55.56 = 200 \times 10^{-4} = 2 \times 10^{-2}$

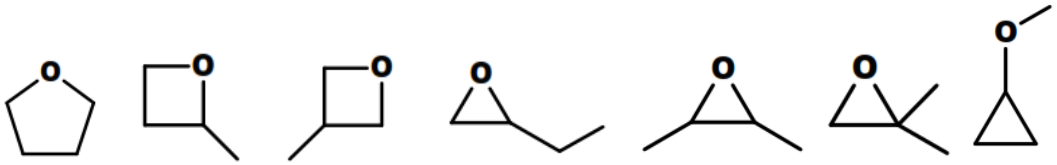
43. Weight of H_2SO_4 consumed $= \left[\frac{1000 \times 1.225 \times 40}{100} - \frac{1000 \times 0.98 \times 20}{100}\right] = 294$ gram

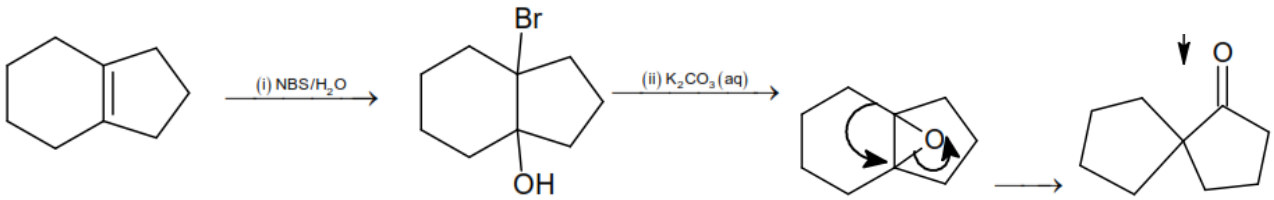
$$\left[\frac{W}{E}\right]_{H_2SO_4} = \frac{Q}{96500}\frac{294}{98} = \frac{Q}{96500}$$

$Q = 96500 \times 3 = 289500$ columb $= \frac{289500}{3600}$ amp. hours $= 80.4$, so $x = 5$

44.

45.





46. Except $Al(OH)_3$, all are gives redox reaction

47. $K[PtCl_3(\pi - C_2H_4)]$, $[Fe(CO)_5]$ contains metal carbon bond

48. $C(s) + O_2(g) \rightarrow CO_2(g);\ \Delta H_f^0 = -394$ kJ / mole

49. $C(s) + O_2(g) \rightarrow CO_2(g);\ \Delta H_f^0 = -394$ kJ / mole

$$H_2(g)\frac{1}{2}O_2(g) \rightarrow H_2O;\ \Delta H_f^0 = -286 \text{ kJ / mole}$$

$$C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l)$$

$$\Delta H^0 = \left[\{6(-394) + 3(-286)\} - 50\right] = -3272 \text{ kJ}$$

50. Conceptual

51. Conceptual